

# Zhiyuan Xiao

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## EDUCATION

### University of Rochester

Candidate for BS in Computer Science and BA in Statistics  
Major GPA: 3.95/4.00 (CS); 3.84/4.00(Statistics)

**Cumulative GPA:** 3.94/4.00

Rochester, USA  
Sept. 2023 - Dec. 2026

**Technical Skills:** Python, C, Java, R, SQL, Swift | Linux, Docker, LaTeX | PyTorch, OpenCV, NumPy, CLIP, T2I-Adapter, GPU-accelerated training and inference, Image and video Generation, DINO

## PUBLICATIONS

Yongsheng Yu, Ziyun Zeng, **Zhiyuan Xiao**, Zhenghong Zhou, Hang Hua, Wei Xiong, and Jiebo Luo. "Aurora: Unified Video Editing with a Tool-Using Agent." *arXiv preprint. arXiv:2605.18748*, 2026. <https://arxiv.org/abs/2605.18748>

## RESEARCH EXPERIENCE

### Aurora: Agentic Video Editing Data and Evaluation

Rochester, USA

**Research Assistant** in Prof. Jiebo Luo's team at the Dept. of Computer Science, University of Rochester

Nov.2025 -Present

- Curated 100K+ image and video editing samples across 8 editing operations and 5 data formats, including subject-to-video (S2V) and subject-video-to-video (SV2V)
- Designed a multi-stage quality-control pipeline using DINO, CLIP, and VLM filtering, which removed >60% of low-quality samples and enabled open-source release of the datasets
- Configured Dockerized Linux GPU benchmark environments with consistent library versions and driver configuration to ensure reproducible model evaluation
- Built the evaluation harness that benchmarked 5 models across the 8 operations, producing the results reported in the paper

## INTERNSHIP EXPERIENCE

### Standard Robotics

Shenzhen, China

*Embedded System Engineer Intern*

Jun. -Jul. 2026

- Developed and deployed a CAN/LoRa-based firmware upgrade system for 150 remote power-control devices, reducing start/stop command completion time by 70% (600s to 180s) and increasing single-command success rate from 75% to 100%
- Built 14 automated tests (firmware flashing, functional modules, communication terminals) plus on-site field validation that isolated and fixed a critical factory-floor defect

### Deepano Technology, a leading provider of visual intelligence platform

Shanghai, China

*Embedded System Engineer Intern*

May. - Aug. 2025

- Optimized image format conversion, denoising, and color correction stages for embedded vision devices on the Horizon Robotics Journey 5 (X5) platform
- Applied SIMD optimization and target-hardware profiling to increase processing throughput by 3x (20 to 60 FPS) on X5, while tuning algorithm parameters and HW/SW interfaces for runtime stability

### Definesys.com, low-code iPaaS platform

Shanghai, China

*Software Engineer Intern (Java)*

Jun. - Jul. 2024

- Developed and daily maintained ETL tools for efficient data extraction, transformation, and loading between cloud servers and joined in the process of localization of the iPaaS Platform by validating 20 functions in each of 4 localized iPaaS releases
- Conducted functionality and performance testing for new platform versions to identify issues and improve system reliability

## SELECTED ACADEMIC RESEARCH

### Multimodal Detection of AI-Generated Twitter Posts | *Python, CLIP, PyTorch*, Team Leader

Oct. -Dec. 2025

- Led a two-member team in building a CLIP-based multimodal detection system for analyzing AI-generated Twitter posts by modeling image-text semantic alignment, independently curated a multimodal dataset of Twitter posts with systematic ablation studies on human- and AI-generated image-text pairs, and co-authored a 10-page research paper presenting the dataset construction, CLIP-based methodology, experimental evaluation, and analysis of multimodal generation artifacts

### Sketch2Anime: AI-Generated Twitter Detection | *Python, PyTorch, OpenCV, NumPy*, Team Member

Apr. -May. 2025

- Developed a two-stage AI system (Sketch2Anime) that transformed simple hand-drawn line sketches into high-quality anime-style illustrations, and combined Canny edge detection with a fine-tuned T2I-Adapter that successfully handled backgrounds, diverse poses, and five artistic styles while preserving the artist's original sketch structure

### Dynamic Storage Allocator (Malloc Lab)— Course Project | *C*, Team Member

Apr. -May. 2025

- Implemented a custom malloc/free/realloc package in C with 8-byte alignment, correct heap invariants, and libc-consistent semantics, achieving 94% space utilization; included a heap-consistency checker for validation

### Statistical Modeling and Inference Project | *R*, Individual Research

Mar. -May. 2024

- Analyzed ten-year (2013-2022) BRFSS obesity data across U.S. states using ANOVA with Dunnett's post hoc test and Kruskal-Wallis tests, and stepwisely a log-linear regression model to identify demographic disparities and quantify annual obesity trends, including a post-2020 increase